

PLOVDIV, Bulgaria

WMO: 156250

Lat: 42.068N

Lon: 24.851E

Elev: 182

StdP: 99.16

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.0	-7.2	-13.1	1.2	-7.1	-10.9	1.5	-4.5	13.7	4.7	12.0	3.7	1.8	280	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	34.9	20.9	33.2	20.5	31.9	20.1	22.8	31.1	21.9	29.8	21.1	28.8	3.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.2	25.4	19.2	14.3	24.5	18.8	13.9	24.2	68.2	31.0	64.9	29.7	62.1	29.0	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.7	8.2	DB	-13.9	38.0	3.7	2.7	-16.6	39.9	-18.8	41.5	-20.9	43.0	-23.6	45.0	(4)
(5)			WB	-14.3	25.2	3.7	1.3	-16.9	26.1	-19.0	26.9	-21.1	27.7	-23.7	28.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	1.2	3.7	8.0	12.7	17.6	21.9	24.4	24.2	19.5	13.6	7.9	2.5	(6)
(7)		DBStd	8.89	4.49	4.69	4.07	3.42	3.01	2.96	2.49	2.40	3.19	3.70	3.97	4.49	(7)
(8)		HDD10.0	887	273	179	88	13	0	0	0	0	12	86	236	(8)	
(9)		HDD18.3	2471	531	409	322	172	49	6	0	0	24	152	314	492	(9)
(10)		CDD10.0	2036	1	3	25	93	237	357	446	442	286	122	22	2	(10)
(11)		CDD18.3	578	0	0	0	2	28	113	188	183	59	4	1	0	(11)
(12)		CDH23.3	6266	0	0	4	44	336	1156	2063	1998	596	66	3	0	(12)
(13)		CDH26.7	2627	0	0	1	6	72	455	964	922	198	9	0	0	(13)
(14)	Wind	WSAvg	2.8	2.7	3.0	3.3	3.1	3.0	3.1	2.8	2.7	2.4	2.4	2.6	(14)	
(15)	Precipitation	PrecAvg	615	48	43	49	44	68	59	58	42	46	40	68	(15)	
(16)		PrecMax	867	109	115	139	81	173	120	170	120	128	112	127	132	(16)
(17)		PrecMin	353	8	3	4	13	20	21	0	0	9	4	2	22	(17)
(18)		PrecStd	130	26	27	30	23	38	28	45	32	31	27	29	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.9	19.1	23.1	27.2	30.9	35.1	38.2	37.1	33.4	28.0	23.0	17.1	(19)
(20)			MCWB	9.1	10.7	12.7	16.5	19.5	21.2	20.8	20.7	19.4	17.8	13.9	10.9	(20)
(21)		2%	DB	12.1	16.2	20.8	24.2	28.8	33.0	35.1	34.9	31.1	25.1	19.1	13.1	(21)
(22)			MCWB	7.5	9.8	11.9	15.1	18.2	21.0	21.0	20.6	18.6	16.4	13.2	9.0	(22)
(23)		5%	DB	9.6	13.8	18.1	22.0	26.9	31.1	33.2	33.1	29.0	22.9	16.2	10.2	(23)
(24)			MCWB	6.7	8.8	10.8	13.9	17.7	20.1	20.6	20.5	18.2	15.5	12.1	7.5	(24)
(25)		10%	DB	7.2	10.9	15.8	20.0	25.1	29.2	31.8	31.7	26.9	20.3	14.0	8.8	(25)
(26)			MCWB	5.2	7.1	9.8	13.2	17.0	19.5	20.2	20.0	17.6	14.5	10.8	6.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	12.4	14.3	18.1	21.1	23.8	23.7	24.0	21.2	19.0	15.7	12.2	(27)
(28)			MCD B	14.4	17.0	20.4	24.3	28.0	31.4	32.4	31.5	28.0	25.3	20.3	15.1	(28)
(29)		2%	WB	8.3	10.4	12.5	16.2	19.7	22.3	22.6	22.8	20.1	17.4	13.9	9.7	(29)
(30)			MCD B	11.0	14.8	18.7	22.1	26.2	30.2	30.8	31.3	27.3	22.9	18.2	11.9	(30)
(31)		5%	WB	6.9	9.1	11.3	14.9	18.6	21.3	21.8	21.7	19.2	16.2	12.7	8.3	(31)
(32)			MCD B	8.8	13.0	17.0	20.5	24.9	28.5	29.7	29.6	26.1	20.8	15.7	10.1	(32)
(33)		10%	WB	5.4	7.6	10.1	13.7	17.6	20.4	21.1	20.8	18.4	15.2	11.4	6.9	(33)
(34)			MCD B	7.0	10.2	14.9	18.8	23.4	27.3	29.0	28.6	24.8	19.3	13.6	8.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.9	9.5	11.1	12.0	12.3	12.7	13.1	13.5	12.5	10.9	9.0	7.8	(35)
(36)			MCDBR	12.1	14.8	16.3	15.8	15.6	15.6	16.2	16.1	16.2	15.2	13.2	11.3	(36)
(37)			MCWBR	8.4	9.5	8.9	8.3	7.4	6.2	6.0	6.1	6.9	7.8	7.9	7.8	(37)
(38)		5% WB	MCD BR	11.0	13.7	14.8	14.0	13.8	14.1	14.2	14.0	13.5	12.9	11.2	9.9	(38)
(39)			MCWBR	8.1	9.1	8.5	8.0	7.2	6.3	5.9	6.1	6.5	7.3	7.1	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.338	0.373	0.425	0.489	0.475	0.478	0.477	0.486	0.457	0.408	0.368	0.343	(40)	
(41)		taud	2.338	2.235	2.117	1.954	2.070	2.107	2.123	2.090	2.132	2.251	2.322	2.329	(41)	
(42)		Ebn at Noon	790	814	810	784	808	805	801	779	770	765	745	745	(42)	
(43)		Edh at Noon	85	111	140	177	162	156	153	153	137	108	87	80	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.68	2.48	3.51	4.57	5.58	6.12	6.43	5.86	4.36	2.94	1.92	1.46	(44)	
(45)		RadStd	0.23	0.31	0.32	0.51	0.42	0.51	0.48	0.40	0.33	0.32	0.19	0.21	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.49	N/A	N/A	N/A	N/A	N/A	-106	-152	N/A	N/A	(46)
(47)	Regional (28 neighbors)	+0.45	N/A	N/A	N/A	+0.61	+0.63	-105	-160	+71	N/A	(47)

Nomenclature: See separate page